The Geography of Careers

Laurits Rømer Hjorth

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Abstract

This paper studies job search across geography and occupations. Using matched employer–employee data for Denmark (1995–2019), I document that worker flows decline sharply with both geographic and occupational distance. I extend a spatial search framework by introducing an occupational dimension: job offers are characterized by a triple of location, occupation, and wage, and offer arrival rates decline with distance in each dimension. I estimate a structural model of on- and off-the-job search that separates the role of offer arrival rates from workers' acceptance decisions. In the model, frictions limit workers' mobility across labor market segments. I use the framework to evaluate how these frictions shape unemployment rates and wage growth across local labor markets.

1 Introduction

The search-and-matching literature emphasizes that job search is a time-consuming process and that the efficiency with which workers and vacancies meet varies across labor markets. In reduced-form models, matching efficiency is typically summarized by a single parameter. In this paper, I provide a structural interpretation of the matching efficiency by decomposing it into two distinct types of frictions: geographic frictions that limit transitions across locations and occupational frictions that limit transitions across occupations.

Local labor markets differ in their occupational composition and therefore in the career prospects they offer. The importance of this heterogeneity for mobility decisions depends on how easily workers can transition between labor market segments. If geographic mobility is easy, workers may accept jobs in regions with limited long-term prospects, knowing that they can relocate later. Similarly, if occupational mobility is frictionless, workers may be less concerned about entering occupations with limited advancement opportunities. When transitions are frictional, however, current job choices have stronger implications for future career paths.

In that way, frictions introduce a dynamic element to job search. Workers do not evaluate job offers in isolation; rather, they consider how accepting a job today affects future job opportunities and expected career outcomes. Similarly, a local labor market offering an attractive wage at a given point in time may

still fail to attract workers if it provides limited scope for upward mobility in the longer run.

From a policy perspective, these frictions are important for regional development strategies. Policies aimed at attracting workers and firms to lagging regions may be ineffective if local labor markets lack viable occupational ladders. When mobility frictions are strong, workers may be reluctant to accept jobs in regions that do not offer long-term career prospects. Ignoring the occupational structure of local labor markets can therefore undermine policies intended to stimulate regional growth.

I begin by documenting that local labor markets offer heterogeneous career prospects across occupations. Estimating a Mincer wage regression with location–occupation interaction fixed effects, I show that the wage premium associated with working in Denmark's densest labor market, Copenhagen, is substantially larger for high-paying occupations such as managers and professionals than for workers in elementary occupations. This implies that the return to geographic mobility depends on a worker's occupation, and that workers in different occupations face different incentives when evaluating job offers from distant labor markets.

I then document that worker flows are highly structured in spatial–occupational space. Most transitions occur within the same segment, and flows across segments follow a gravity pattern with steep distance gradients in both dimensions. The gravity estimates indicate that crossing a provincial border reduces worker flows by around 34%, while switching 1-digit occupation reduces flows by around 90%. These patterns are consistent with prior evidence that workers are less likely to match with jobs that are geographically distant (Schmutz and Sidibé, 2019) or occupationally distant (Bocquet, 2024).

To analyze these frictions affect workers' career decisions, I develop a structural model of job search across geographic and occupational space. Building on the spatial job search framework of Schmutz and Sidibé (2019), I introduce an occupational dimension such that job offers are characterized by a triple of location, occupation, and wage. Workers search both on-the-job and off-the-job, and offer arrival rates decline with distance in both dimensions.

These frictions operate through two mechanisms. First, job offers arrive less frequently from distant segments, reflecting search barriers and limited information flows. Second, workers incur mobility costs when transitioning across locations or occupations, such as relocation or retraining costs. As a result, workers may optimally reject job offers even when they offer higher wages, departing from the canonical Burdett–Mortensen framework in which workers always accept higher-paying offers. In this setting, job-to-job transitions occur only when wage gains are large enough to compensate for one-time mobility costs.

When an offer arrives, workers decide whether to accept or reject it, taking into account current utility, mobility costs, and how the offer unlock future job opportunities. The structural framework allows me to disentangle two sources of variation in worker mobility: differences in offer arrival rates and differences in acceptance decisions. I use the model to quantify how spatial and occupational frictions shape unemployment rates, wage growth, and regional disparities in

labor market outcomes.

Finally, the findings have direct policy relevance. In Denmark, policies such as the *Better Balance* reforms have sought to relocate public-sector employment to lagging regions. The results suggest that regional policy must account not only for the number of jobs in a region, but also for the occupational ladders embedded within local labor markets. Unequal occupational structures across space contribute to persistent regional disparities, highlighting the importance of career opportunities within places.

2 Related literature

A large literature models unemployment, job creation, and wage determination using search-and-matching frameworks. Canonical contributions by Mortensen and Pissarides (1994) provide the baseline for understanding how job offers arrive and matches form. Extensions that incorporate on-the-job search emphasize the importance of job-to-job transitions for wage dynamics and worker reallocation (e.g., Burdett and Mortensen, 1998; Cahuc et al., 2006). In most of these models, matching efficiency is treated as a reduced-form parameter. A growing strand of work seeks to interpret matching frictions structurally—for example, by modeling the geography of offers (Marinescu and Rathelot, 2018). By introducing an explicit occupational dimension into a spatial search framework, this paper contributes to this literature by decomposing matching frictions into geographically and occupationally driven components.

A related literature documents large and persistent geographic heterogeneity in labor market outcomes and analyzes how workers migrate in search of better job opportunities, where Kennan and Walker (2011) is an early contribution. Empirical studies document steep distance gradients in workers' search intensity (Manning and Petrongolo, 2017). Schmutz and Sidibé (2019) develop a spatial search model in which offer arrival rates decline with geographic distance. I extend this framework by introducing an occupational dimension, allowing offer arrival rates to depend jointly on geographic and occupational distance and thereby linking regional labor market structure to career dynamics.

A parallel literature studies occupational choice and career dynamics, emphasizing that transitions between occupations involve adjustment costs and affect lifetime earnings. Seminal contributions model occupational choice over the life cycle and highlight the role of human capital accumulation (Keane and Wolpin, 1997). Empirical work documents the persistence of occupation-specific human capital and the costs of occupational switching (Kambourov and Manovskii, 2009). More recent studies construct occupational networks based on skill similarity to analyze feasible career transitions and worker adjustment to structural change (Bocquet, 2024). This paper connects these insights to spatial search by using skill-based occupational distances (derived from ESCO) to parameterize occupational frictions and by estimating how occupational and geographic frictions jointly shape mobility and wage trajectories.

Solving dynamic job search models with multidimensional state spaces is

computationally demanding. This paper draws on methods developed in the dynamic discrete choice literature (Rust, 1987; Hotz and Miller, 1993; Arcidiacono and Miller, 2011) to make estimation tractable. Following Arcidiacono et al. (2023), I use the conditional choice probability (CCP) approach to recover acceptance probabilities and solve the dynamic programming problem without repeatedly computing the full value function at each parameter guess. This approach allows the model to remain computationally feasible despite the large number of spatial–occupational segments.

Finally, this paper relates to the literature on regional policy and agglomeration. Work on agglomeration and market size (e.g., Moretti and Yi, 2024; Duranton and Puga, 2004) emphasizes how geography shapes the distribution of jobs and career opportunities. Structural models that quantify the channels through which geography and occupational composition affect reemployment and wage growth are essential for evaluating regional development policies. By structurally decomposing spatial and occupational frictions, this paper provides tools for assessing how regional labor market structure influences adjustment and inequality.

3 Data

The analysis uses Danish matched employer–employee administrative data from the IDA database covering the period 2005–2019. I observe the entire Danish population and their employment status each year in November. For employed workers, I observe their primary job, including occupation, establishment location, and hourly wage. This structure allows me to follow individuals over time, observe job-to-job (EE), employment-to-unemployment (EU), and unemployment-to-employment (UE) transitions, and distinguish whether these transitions occur within or across firms, occupations, and localities.

Because unemployed workers are not directly assigned to an occupation or work location, I impute these characteristics using the worker's most recent employment spell. This allows workers to be continuously tracked in location-occupational space and ensures that mobility can be measured symmetrically for EE and UE transitions.

I restrict the sample to employed workers aged 20–60, but I only allow individuals to enter the sample after having reached their highest level of education.

Each worker-year observation is assigned to one of J geographic locations and O occupations, enabling the construction of complete employment histories in spatial–occupational space. I use Danish provinces to represent local labor markets ($J = 10$), and occupations are classified using 1-digit ISCO codes ($O = 9$). This gives a total of 90 labor market segments.

Regarding the spatial aggregation, it is important to choose a level that, on the one hand, reveal variation in labor market composition and conditions and, on the other hand, defines locations that are sufficiently large to constitute economically distinct labor markets from the worker's perspective. In the data, only a relatively small share of workers commute across provincial borders (5%),

indicating that provinces are large enough to be treated as separate local labor markets in which job location meaningfully affects worker outcomes. With respect to occupational aggregation, I found that the 1-digit ISCO classification is the most detailed level I can use without getting excessively thin labor market segments with very few observations.

As pointed out by Shimer (2012), the annual structure of the data raises a concern about the time aggregation problem. Because workers are observed only once per year, transitions that occur within the year may be missed. For example, a worker who transitions from employment to unemployment and back to employment within a single calendar year will appear as continuously employed in the November snapshots, leading to an underestimation of hazard rates for employment-to-unemployment (EU) and unemployment-to-employment (UE) transitions. To mitigate some of this concern, I exploit supplementary administrative records (IND) on unemployment benefits. These records allow me to identify whether a worker was registered as unemployed at any point during the year, even if they were employed in both the November observation of year t and year $t-1$. Workers for whom this is the case are classified as having experienced both an EU and UE transition during year t. This within-year unemployment indicator restores a large share of the unemployment transitions that would otherwise be missed in the November snapshots, reducing the downward bias in estimated EU and UE hazard rates that would arise from relying on the annual panel alone.

Distances between labor market segments are an essential part of the analysis. I construct two distance matrices, one for geographical distances and one for occupational. The geographical distances are measured as the driving distance between province centriods, while the occupational distances are based on the skill and knowledge composition of each occupation. This information is obtained from ESCO [1]. Each occupation o is represented by a vector of K standardized skill and knowledge requirements, $s_o = (s_{o1}, \ldots, s_{oK})$. I use these vectors to measure the occupational distance as the cosine disimilarity:

$$D^{\text{occ}}_{oo'} = 1 - \frac{s_o \cdot s_{o'}}{\|s_o\| \|s_{o'}\|}, \tag{1}$$

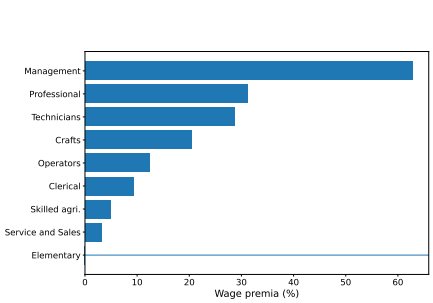
This measure ranges from 0 for identical occupations to 2 for occupations that require entirely distinct skill and knowledge sets. In robustness checks, I also consider L_1 and L_2 distance measures, as well as variance-weighted versions that account for differences in the relative importance of specific skills. An important motivation for the structural model is that career prospects vary across occupations and locations jointly: the wage premium associated with working in a given location is not uniform across occupations, but depends on the type of work. To document this heterogeneity, I estimate a Mincer wage regression including standard worker controls and location–occupation interactions:

$$\log w_i = \alpha + \beta_{o(i)} + \delta_{j(i),\, o(i)} + \mathbf{X}_i'\theta + \varepsilon_i, \tag{2}$$

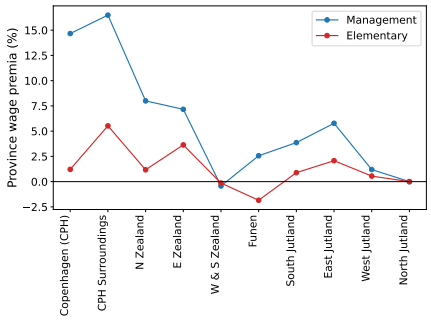
[1] European Skills, Competences, Qualifications and Occupations

where w_i is the hourly wage of worker i, $\beta_{o(i)}$ are occupation fixed effects, $\delta_{j(i),\,o(i)}$ are location–occupation interaction fixed effects, $\mathbf{X}_i$ is a vector of worker characteristics including age, work experience, and field and level of education, and ε_i is an error term.

Figure 1(a) plots the estimated occupation fixed effects, $\hat{\beta}_o$, which show that managers earn the highest wage premium, around 60% above workers in Elementary occupations. Figure 1(b) shows that this premium is not homogeneous across locations. In particular, managers gain a substantial higher wage premium when employed in the most dense labor market, Copenhagen, while workers in Elementary occupations earn a similar wage across all locations. This pattern suggests that the return to geographic mobility depends strongly on a worker's occupation, motivating a model that treats geography and occupation as jointly determined dimensions of job search.



(a) Occupation wage premiums



(b) Location wage premiums per occupation

Figure 1: Location–occupation wage premiums estimated from equation (2). Coefficients in panel (a) is relative to the Elementary occupations and to the North Jutland in panel (b)

Another important motivation for modelling workers' search across geography and occupation jointly is that these dimensions of mobility is actually relevant when looking at labor flows. Table 1 shows that a significant portion of labor market transitions happens across segments, with roughly two-thirds of employment-to-employment (EE) transitions occur within the same labor market segment. Among workers who move across segments, mobility is primarily geographic.

Workers reentering employment from unemployment (UE) are even more mobile, where 44 % find a job in a segment different from where they were most recently employed. Unemployed workers are particularly more likely to move to another occupation compared to employed workers, while the geographic mobility are similar for the two groups.

It is also worth noting that also workers who do not change employer (Stay) can make occupational moves which happens for 8.4% of the workers who stay

with the same employer. This is not a margin of adjustment that I model in the structural framework. In Arcidiacono et al. (2023) these within-employer occupational moves are modelled as exogenous transitions that do not involve a worker decision.

Table 1: Mobility across labor market segments, by transition type, yearly

	Stay (NO)	**EE**	**UE**
No move	89.9%	67.4%	56.1%
Only Spatial move	1.5%	13.4%	7.7%
Only Occupational move	8.4%	8.6%	17.5%
Spatial & occupational move	0.2%	10.6%	18.8%
Total	100%	100%	100%
N (1,000)	17,413	2,026	578

Notes: A *spatial move* is defined as a change in the province of the workplace. A *occupational move* is defined as a change in the two-digit ISCO occupation of the worker's primary job. Provinces correspond to the 10 Danish provinces (*landsdele*), giving $J = 10$ geographic units. Occupations are classified at the two-digit ISCO level, giving $O = 43$ occupational groups. Sample covers employed workers aged 18–60 observed in the Danish matched employer–employee register, 1995–2019. N reports the number of worker-year observations in thousands.

While Table 1 documents whether workers move across segments, it does not characterize which specific origin–destination transitions are more likely to occur. To examine the nature of mobility in spatial–occupational space, I estimate a gravity equation that models bilateral worker flows between origin segment (j, o) and destination segment (j', o'):

$$\begin{aligned} \log F_{j,o}^{j',o'} = \alpha_{j,o} + \nu_{j',o'} + \delta^G \, I\{j \neq j'\} + \tau \, \log D_{j,j'}^G \\ + \delta^O \, I\{o \neq o'\} + \omega \, D_{o,o'}^O + \varepsilon_{j,o}^{j',o'} \end{aligned} \tag{3}$$

Here, $F_{j,o}^{j',o'}$ denotes the flow of workers from segment (j, o) to (j', o'). The fixed effects of origin $\alpha_{j,o}$ capture differences in the size and intensity of the exit of each segment of origin, while the fixed effects of destination $\nu_{j'o'}$ capture the relative attractiveness of the destinations.

The OLS estimates of the gravity equation are reported in Table 2. The estimation is done separately for each transition type. The coefficients δ^G and δ^O capture the average percentage reduction in flows associated with crossing a provincial border or switching occupations, respectively. These have a significant impact on flows, e.g. EE transitions drop by about 34% ($\hat{\delta}^G = -0.42$) and 90% ($\hat{\delta}^O = -2.52$), when they happen across province borders and across 1-digit occupational codes respectively. Consistent with the descriptive shares, the coefficients are smaller for UE transitions, indicating that unemployed workers are more likely to move across occupational and geographic borders, either because

Table 2: Gravity estimates of worker flows across spatial and occupational segments

	δ^G	τ	δ^O	ω	R^2
EE	−0.42 (0.01)	−0.68 (0.01)	−2.52 (0.02)	−1.85 (0.02)	0.692
UE	−0.30 (0.01)	−0.63 (0.01)	−1.86 (0.02)	−1.56 (0.02)	0.656

Notes: Results from estimating (3) with pooled OLS. Standard errors in parentheses.

they search more intensively across segments and therefore experience lower frictions or because unemployed workers are more willing to pay the shifting costs because their outside option – of staying unemployed – is worse.

The coefficients τ and ω capture the elasticity of flows with respect to geographic and occupational distance, respectively. These findings are consistent with prior work on interregional (Schmutz and Sidibé, 2019) and interoccupational (Bocquet, 2024) mobility.

Taken together, the descriptive shares and gravity estimates show that worker mobility is systematically structured by both geographic and occupational distance. Because observed flows conflate offer arrival rates and worker acceptance decisions, a structural model is required to disentangle these mechanisms - this is the goal of the next section.

4 Model

The model builds on the wage posting framework in Burdett and Mortensen (1998) extended to a segmented labor market where jobs differ along three dimensions: location, occupation, and wage. Workers conduct both off-the-job and on-the-job search. Workers can only move to a labor market segment by accepting a job offer from that segment.

The model is dynamic. Workers make accept or reject offers considering the choice's impact on current utility and how it affects the arrival of future job opportunities.

The model is intended to highlight the nature of workers' search strategies along multiple dimensions. On the other hand, it abstract from broader equilibrium mechanisms on the housing market that might be affected by workers' migration decisions and it also do not cover the strategic interaction between workers' job search and firms' wage posting behavior. Instead, the wage offer distributions are treated as exogenously given by segment-specific distributions $F_{j,o}(w)$.

4.1 A segmented labor market

The economy consists of J geographic locations indexed $j \in \mathcal{J}$ and O occupations indexed by $o \in \mathcal{O}$. Each pair, (j, o), defines a distinct segment of the labor market, so that the economy can be viewed as a collection of $J \times O$ submarkets. Within a segment, jobs are characterized by a wage w drawn from a segment-specific distribution $F_{j,o}(w)$ with common support $[\underline{w}, \overline{w}]$.

4.2 Workers

The employment status of workers is denoted i and workers can be either employed ($i = e$) or unemployed ($i = u$). Employed workers are associated to a specific labor market segment (j,o) given by their workplace, while unemployed workers belong to the segment of their most recent workplace.

The per-period utility of an employed worker in (j,o) is given by

$$u^e_{j,o}(w) = \gamma_j + \eta^e_o + \mu(w), \tag{4}$$

which consist of location-specific amenities, γ_j, the nonpecuniary utility of working in occupation o, η^e_o, and the pecuniary utility of wage, $\mu(w)$. Similarly, the flow utility of being unemployed in segment (j,o) is

$$u^u_j = \gamma_j + \eta^u + \mu(b), \tag{5}$$

, where η^u is the nonpecuniary value of being unemployed and b is the unemployment insurance.

4.3 Job offer arrival rates

Job offers arrive according to a Poisson process. The arrival rate is controlled by three parameters: the segment-specific arrival rate, $\lambda^i_{j',o'}$, that captures the job creation intensity of the destination segment (j', o'). That is, $\lambda^i_{j',o'}$ measures how frequently new job openings arise in segment (j', o'), independent of where workers are currently located.

A worker in segment (j, o) do not observe all such openings, but only a fraction that depends on the spatial and occupational frictions between the worker's current segment (j, o) and the destination (j', o'). Let $\tau^i_{j,j'}$ denote the spatial accessibility between locations j and j', and $\omega^i_{o,o'}$ the occupational accessibility between occupations o and o'.

Combining these three parameters give the effective arrival rate of offers from destination segment (j', o') for a worker currently employed in (j, o) by

$$\Lambda^e_{(j,o)\to(j',o')} = \tau^e_{j,j'}\, \omega^e_{o,o'}\, \lambda^e_{j',o'}. \tag{6}$$

Analogously, for unemployed workers searching for a job from a segment (j, o), where they were most recently employed,

$$\Lambda^u_{(j,o)\to(j',o')} = \tau^u_{j,j'}\, \omega^u_{o,o'}\, \lambda^u_{j',o'}. \tag{7}$$

The effective arrival rates (6 and 7) parallels a gravity pattern of worker flow, in which observed mobility depends on both the mass (job opportunities) of the destination and the distance (frictions) between origin and destination. The accessibility parameters are allowed to vary between employed and unemployed workers which is a possible explanation for the differences in mobility across the two groups documented by the gravity equations.

A key feature of the model is that the arrival rates of offers depend on the network of occupations and locations such that workers must be strategic when deciding whether to accept or reject an offer since it has implications for the which job offers they will receive in the future.

Upon accepting an offer (j', o'), the worker incurs a one-time mobility cost, $c^i_{(j,o)\to(j',o')} = \kappa^i_{jj'} + \phi^i_{oo'}$, which depends on the transition of location and occupation, respectively.

4.4 Value functions

The model is set in continuous time and the value of being employed in location, j, occupation o and with wage w is given by

$$\begin{aligned}\rho V^e_{j,o}(w) = u^e_{j,o}(w) + \sum_{j'\in\mathcal{J}}\sum_{o'\in\mathcal{O}} \tau_{j,j'}\,\omega_{o,o'}\,\lambda_{j',o'} \\ \int \mathbb{E}\Big[\max\Big\{V^e_{j,o}(w) + \varepsilon_r,\; V^e_{j',o'}(w') - c^e_{(j,o)\to(j',o')} + \varepsilon_a\Big\}\Big] - V^e_{j,o}(w) dF_{j',o'}(w') \\ + \delta_{j,o}\big(V^u_{j,o} - V^e_{j,o}\big)\end{aligned} \tag{8}$$

This function captures the expected utility of being employed in a given segment (j, o) with a wage w, which consists of the current utility flow, the expected value from receiving job offers, and the risk of becoming unemployed. The expected value of job offers depends on how frequently they occur which depends on the location of the worker as given by (6), the distribution of offered wages, and the shifting costs $(c^e_{(j,o)\to(j',o')})$.

The preference shocks associated to the two options of rejecting or accepting, ε_r and ε_a, are drawn from the extreme value type-I distribution with scale parameter of 1.

The advantage of formulating the model in continuous time is that I do not have to consider the scenario where more than one offer arrives within the same period. Instead each choice situation is binary and is related to non-negative expected value according to the Bellman equation (9). The distrbutional assumption on the preference shocks further simplifies the model by allowing me to replace the discontinuous max operator with a smooth logsum term, which is the expected value under EV type-I shocks. This is a standard approach in the dynamic discrete choice literature (Rust, 1987; Hotz and Miller, 1993; Arcidiacono and Miller, 2011) that makes the model computationally tractable while still capturing the key economic trade-offs in workers' acceptance decisions.

Specifically, the expected value of receiving an offer from segment (j', o') can be expressed as:

$$\mathbb{E}\left[\max\{V^e_{j,o}(w) + \varepsilon_r, V^e_{j',o'}(w') - c^e_{(j,o)\to(j',o')} + \varepsilon_a\}\right] \tag{9}$$

$$= \log\left(\exp\left(V^e_{j,o}(w)\right) + \exp\left(V^e_{j',o'}(w') - c^e_{(j,o)\to(j',o')}\right)\right), \tag{10}$$

which shows that the expected value of any offer is a continuous and strictly positive function of the offers' observable characteristics. The economic intuition behind this smoothing is that it captures the idea that workers may have some idiosyncratic preference for accepting or rejecting offers, and that even if an offer is not strictly better than the current job in terms of observable characteristics, it may still have some positive expected utility due to unobserved factors.

The value of being unemployed follows the same structure as (9), but without the risk of becoming unemployed:

$$\rho V^u_{j,o} = u^u_{j,o} + \sum_{j'\in\mathcal{J}}\sum_{o'\in\mathcal{O}} \tau^u_{j,j'}\,\omega^u_{o,o'}\,\lambda^u_{j',o'} \int \mathbb{E}\Big[\max\left\{V^u_{j,o} + \varepsilon_r,\ V^e_{j',o'}(w') - c^u_{(j,o)\to(j',o')} + \varepsilon_a\right\}\Big]\, dF_{j',o'}(w') \tag{11}$$

5 Estimation

The structural model introduced in Section 4 has three sets of parameters: the segment-specific wage offer distributions $F_{j,o}(w)$, the offer arrival rates $\lambda_{j,o}$, and the friction and preference parameters $\theta = (\tau, \omega, \kappa, \varphi, \gamma, \eta, \mu)$.

The challenges related to the identification and estimation of these parameters are threefold. First, and related to data, the model includes a large number of segments defined by locations and occupations. Even with the large sample size of the Danish administrative data, some labor market transitions between segment pairs occur rarely, making it difficult to estimate transition-specific parameters with precision. This challenge is addressed by aggregating to a relatively high level of segmentation – provinces and 1-digit occupations – which ensures sufficient observations in each origin-destination cell while retaining the key sources of heterogeneity of interest. Second, the model is dynamic and involves forward-looking behavior, as workers' job acceptance decisions depend on the entire distribution of future career opportunities rather than just current wages and amenities. This creates a computational challenge for estimation, since evaluating the likelihood requires solving for the value function at each candidate parameter vector. Third, observed worker flows conflate two distinct mechanisms: the rate at which job offers arrive from a destination segment, and workers' decisions to accept or reject those offers. This means that observed hazard rates can not be directly interpreted as conditional choice probabilities of accepting offers which complicates the identification of the model's parameters.

I discretize wages into ten bins such that the hazard rates can be measured for a group of workers within the same segment and with a similar wages. The hazard rate of a worker currently employed in segment (j, o) at wage w transitioning to a job in segment (j', o') at wage w' is given by:

$$h_{j,o,w}^{j',o',w'} = \tau_{j,j'} \cdot \omega_{o,o'} \cdot \lambda_{j',o'} \cdot f_{j',o'}(w') \cdot p_{j,o}^{j',o'}(w, w'), \tag{12}$$

where $f_{j',o'}(w')$ is the probability mass of offers within wage bin w' from segment (j', o') according to the offer distribution $F_{j',o'}(w)$, and $p_{j,o}^{j',o'}(w, w')$ is the conditional choice probability (CCP) of accepting the offer (j',o',w') conditional on being employed in segment (j, o) at wage w.

The hazard rates clearly show that hazards rates do not directly correspond to the model's conditional choice probabilities. The identification strategy that allow me estimate the structural parameters from observed hazards follows the approach presented in Arcidiacono et al. (2023), adapted to the spatial-occupational setting of this paper. The key insight is that the model's structure implies a mapping from the observed hazard rates to the underlying parameters, which can be exploited for estimation. The estimation proceeds in two steps. First, I recover the continuous-time hazard rates of transitioning between segments from the observed annual transition probabilities in the data, while accounting for competing risks. Second, I use these hazard rates to estimate the model's parameters by matching the predicted hazard rates from the model to those recovered from the data.

Recovering continuous-time hazard rates from annual data. The data provide annual snapshots of workers' labor market status, whereas the model is formulated in continuous time. This naturally raises a concern about whether I can obtain the continuous-time hazard rates required for estimation from the observed annual transition probabilities. The time aggregation problem is well-known in the literature on labor market flows, and it arises because discrete observation of an inherently continuous process leads to systematic underrecording of transitions that occur between observation points Shimer (2012).

On average, around 20 % of workers transition into a new job within a year. If each worker transition according to a continous-time Poisson proccess, the amount of workers that experience more than one transition within a year is given by $1 - e^{-\lambda} - \lambda e^{-\lambda}$, where λ is the average number of transitions per year. With $\lambda = 0.2$, this implies that around 2% of workers experience more than one transition within a year, which would be missed in the annual snapshots.

A particular concern is the missing transitions into unemployment (EU) which will not be captured in the yearly observations if the worker finds a new job within the year – in that case the transition will look as a job-to-job (EE). For this particular issue, I exploit the data on unemployment benefits to recover both the EU and EU transition as discussed in section 3.

I estimate the hazard rates by first calculating the total hazard of transitioning away from any segment-wage cell:

$$h_{j,o,w}^{move} = -\log(P_{j,o,w}^{Stay}), \tag{13}$$

where $P^{Stay}_{j,o,w}$ is the observed annual probability of remaining in the same segment and wage bin, such that $h^{move}_{j,o,w}$ represents the total hazard of leaving the cell (j,o,w). The hazard of transitioning to any specific job (j',o',w') is then given by the product of the total hazard and the observed share of transitions from (j,o,w) to (j',o',w') among all transitions out of (j,o,w):

$$h^{j',o',w'}_{j,o,w} = h^{move}_{j,o,w} \cdot \frac{P^{j',o',w'}_{j,o,w}}{1 - P^{Stay}_{j,o,w}}, \tag{14}$$

which defines the observed hazard rates which are used for estimation. This approach accounts for the competing risks of transitioning to different segments and wage bins.

Estimating wage offer distributions To identify the offer distribution, I focus on within-segment transitions, and I use the formulation from Arcidiacono et al. (2023) to isolate the wage offer distribution $f_{j',o'}(w)$ from hazard rates:

$$\frac{h^{j,o,w}_{j,o,w}}{\sum_x h^{j,o,x}_{j,o,x}} = \frac{\tau_{j,j} \cdot \omega_{o,o} \cdot \lambda_{j,o} \cdot f_{j,o}(w) \cdot p^{j,o}_{j,o}(w,w)}{\tau_{j,j} \cdot \omega_{o,o} \cdot \lambda_{j,o} \cdot \sum_x f_{j,o}(x) \cdot p^{j,o}_{j,o}(x,x)} = \hat{f}_{j,o}, \tag{15}$$

which uses that the frictions cancel out because all transitions are happening within the same segment. The acceptance probability also cancels out since all choice probabilities in (15). Finally and per definition, the summation over the wage density sums to one.

Equation (15) allows me to recover a non-parametric estimator for the wage offer distribution. To smooth the wage offer distribution across wage bins, I fit a segment-specific beta distribution on top of the nonparametric estimator. Concretely, for each segment (j,o), I choose parameters $\alpha_{j,o}$ and $\beta_{j,o}$ to minimize the Kullback-Leibler divergence from the nonparametric density $\hat{f}_{j,o}$ to the beta density $f_{\text{Beta}}(\cdot\,; \alpha_{j,o}, \beta_{j,o})$:

$$(\hat{\alpha}_{j,o}, \hat{\beta}_{j,o}) = \underset{s\alpha_{j,o},\, \beta_{j,o}>0}{\arg\min}\; D_{\text{KL}}\Big(\hat{f}_{j,o} \parallel f_{\text{Beta}}(\cdot\,; \alpha_{j,o}, \beta_{j,o})\Big) \tag{16}$$

$$= \underset{\alpha_{j,o},\, \beta_{j,o}>0}{\arg\min} \sum_b \hat{f}_{j,o}(b) \ln \frac{\hat{f}_{j,o}(b)}{f_{\text{Beta}}(b;\, \alpha_{j,o}, \beta_{j,o})} \tag{17}$$

where the sum runs over wage bins b and where the density of each bin in the beta distribution is calculated as $f_{\text{Beta}}(b; \alpha_{j,o}, \beta_{j,o}) = F_{\text{Beta}}(b; \alpha_{j,o}, \beta_{j,o}) - F_{\text{Beta}}(b-1; \alpha_{j,o}, \beta_{j,o})$. This procedure yields a smoothed estimate of the wage offer distribution for each segment, which is used in the subsequent estimation of the model's parameters.

Maximum Likelihood Estimation With the wage offer distributions, $f_{\text{Beta}}(w; \alpha_{j,o}, \beta_{j,o})$, recovered as described above, I estimate the remaining structural parameters – shifting costs, search frictions, non-pecuniary utility of working in each location and occupation – by maximum likelihood.

A key question is whether the search frictions ($\tau_{j,j'}$ and $\omega_{o,o'}$) and mobility costs ($\kappa_{j,j'}$ and $\phi_{o,o'}$) are seperately identified. I argue that these structural parameters have distinct effects on hazard rates and therefore are identified from the data.

The transition hazard from segment (j, o) to (j', o) for a worker at wage w accepting offer w' is:

$$h_{j,o,w}^{j',o',w'} = \tau_{j,j'} \cdot \lambda_{j',o'} \cdot f_{j',o'}(w') \cdot \frac{\exp\big(V_{j',o}(w') - \kappa_{j,j'}\big)}{\exp(V_{j,o}(w)) + \exp\big(V_{j',o}(w') - \kappa_{j,j'}\big)}, \quad (18)$$

where I have inserted the closed-form expression for the CCP which follows from the assumption of preference shocks following an EV1 distribition. I disregard the occupational frictions and moving costs (18)because the hazard is related to a within-occupation transition.

Because of the non-linarity of the CCP, the moving costs $\kappa_{j,j'}$ have a non-uniform effect across wage bins, where an increase in mobility costs have the larger affect for workers where the wage offer from (j',o) is close to the indifferrence threshold. By contrast, the search frictions $\tau_{j,j'}$ have a uniform effect across wage bins since they enter the hazard rate as a multiplicative factor that does not interact with the CCP. This means that mobility costs are identified from the shape of the hazard across wage bins, while search frictions are identified from the level of the hazard between locations.

Likelihood function. Write up likelihood function and estimation procedure.

6 Results

WORK IN PROGRESS

7 Discussion

The estimation procedure described in Section 5 assumes that the observed wage distribution within each location-occupation cell is informative about the wage offers available to any worker in that cell. This assumption may be violated if workers are heterogeneous in ways that systematically affect their wages across all jobs. In that case, the relevant wage offer distribution for a given worker is not what other workers earn in a particular job, but what that specific worker would earn there – reflecting their portable human capital rather than the average productivity of the cell. A possible way to address this is to use the Mincer wage regression estimated in Section 3. Specifically, one can extract the estimated worker fixed effects – capturing returns to education, age, and experience – and subtract them from observed wages to obtain residualized wages that reflect only the job-specific wage premium net of worker characteristics. The estimation procedure described in Section 5 then proceeds on these residualized wages rather than raw wages, so that the offered wage distribution captures variation

in job quality rather than variation in worker quality. This concern is less to some extent migitated by the introduction of occupations in the model, because it makes the workers within each segment more homogenous.

The present paper defines worker mobility in terms of job location – the location of the establishment where the worker is employed. This is the natural definition from a labor market perspective, as it determines which local labor market the worker participates in. However, job location and residential location need not coincide, and the distinction between the two opens up a rich set of questions at the intersection of labor and housing markets. The Danish administrative data used in this paper are well-suited to explore this distinction, as they allow tracking of both residential and workplace locations simultaneously for each worker. This data feature creates an opportunity to study commuting patterns, residential sorting, and the joint dynamics of housing and labor market decisions — aspects that are entirely abstracted away in the current framework. The connection between housing and labor markets has long been noted by policymakers and practitioners. A commonly cited argument is that a well-functioning housing market is a precondition for an efficient labor market since it allow workers to easily move towards areas where job opportunities are attractive. To formalize this argument could be a interesting extension of the current model. A way to introduce this into the model would be to calculate wage offer distributions net of commuting costs such that workers make a residential choice while forming expectations of their future job opportunities where wages is substracted the expected commute. This would e.g. capture the trade-off between paying higher rents to live in a city against having low commute time to potential job offers that might arise in the future. Incorporating residential choice would also allow the model to speak to policy questions around housing supply, zoning regulations, and mobility subsidies – questions that are increasingly central to debates about regional inequality and the geographic concentration of economic opportunity.

8 Conclusion

This paper studies worker mobility across geography and occupations and provides a structural interpretation of matching frictions in segmented labor markets. Using matched employer–employee data for Denmark covering 1995–2019, I document that worker flows decline sharply with both geographic and occupational distance.

To interpret these patterns, I develop a structural job search model in which job offers are characterized by a triple of location, occupation, and wage, and in which offer arrival rates decline with distance in both dimensions. A central contribution of the paper is to separate two mechanisms that are confounded in reduced-form analyses — the rate at which job offers arrive from distant segments, and workers' decisions to accept those offers — by exploiting the differential information content of within-segment and cross-segment transitions. The gravity patterns documented in Section 3 reflect both frictions, but their

structural sources are distinct: offer arrival rates are governed by geographic and occupational accessibility, while acceptance decisions additionally depend on wages and one-time mobility costs.

I am currently in the process of estimating the structural model. The structural decomposition has direct implications for how we think about regional labor market policy. A region that attracts few workers may do so because offers from that region rarely reach workers elsewhere — a friction problem — or because workers receive those offers but find them unattractive given mobility costs and the career prospects — an acceptance problem. These two mechanisms call for different policy responses. Reducing geographic frictions, for instance through better information about distant vacancies or reduced commuting costs, raises the rate at which workers are exposed to opportunities in lagging regions. Reducing mobility costs, through relocation subsidies or retraining support, instead raises the probability that workers act on offers they already receive. The framework developed here provides a tractable way to quantify the relative importance of each mechanism and to assess the likely effectiveness of policies targeting either margin.

References

Arcidiacono, Peter, Attila Gyetvai, Arnaud Maurel, and Ekaterina Jardim (2023) "Identification and Estimation of Continuous-Time Job Search Models with Preference Shocks *," technical report.

Arcidiacono, Peter and Robert A. Miller (2011) "Conditional Choice Probability Estimation of Dynamic Discrete Choice Models With Unobserved Heterogeneity," *Econometrica*, 79, 1823–1867, 10.3982/ECTA7743.

Bocquet, Léonard (2024) "CAMBRIDGE WORKING PAPERS IN ECONOMICS JANEWAY INSTITUTE WORKING PAPERS The Network Origin of Slow Labor Reallocation The Network Origin of Slow Labor Reallocation *," technical report, www.econ.cam.ac.uk/cwpewww.janeway.econ.cam.ac.uk/working-papers.

Burdett, Kenneth and Dale T Mortensen (1998) "Wage Differentials, Employer Size, and Unemployment," technical report, https://www.jstor.org/stable/2527292?seq=1&cid=pdf-.

Cahuc, Pierre, Fabien Postel-Vinay, and Jean-Marc Robin (2006) "Wage Bargaining with On-the-Job Search: Theory and Evidence," technical report.

Duranton, Gilles and Diego Puga (2004) "MICRO-FOUNDATIONS OF URBAN AGGLOMERATION ECONOMIES," 10.1016/S0169-7218(04)07048-0.

Hotz, V Joseph and Robert A Miller (1993) "Conditional Choice Probabilities and the Estimation of Dynamic Models," technical report.

Kambourov, Gueorgui and Iourii Manovskii (2009) "Occupational Mobility and Wage Inequality," *Review of Economic Studies*, 76, 731–759, 10.1111/j.1467-937X.2009.00535.x.

Keane, Michael P. and Kenneth I. Wolpin (1997) "The career decisions of young men," *Journal of Political Economy*, 105, 473–522, 10.1086/262080.

Kennan, John and James R. Walker (2011) "The Effect of Expected Income on Individual Migration Decisions," *Econometrica*, 79, 211–251, 10.3982/ECTA4657.

Manning, Alan and Barbara Petrongolo (2017) "How local are labor markets? Evidence from a spatial job search model," *American Economic Review*, 107, 2877–2907, 10.1257/aer.20131026.

Marinescu, Ioana and Roland Rathelot (2018) "American Economic Association Mismatch Unemployment and the Geography of Job Search," *Journal: Macroeconomics*, 10, 42–70, 10.2307/26528444.

Moretti, Enrico and Moises Yi (2024) "Size Matters: Matching Externalities and the Advantages of Large Labor Markets," technical report, National Bureau of Economic Research, 10.3386/w32250.

Mortensen, Dale T and Christopher A Pissarides (1994) "Job Creation and Job Destruction in the Theory of Unemployment," technical report, https://academic.oup.com/restud/article/61/3/397/1589192.

Rust, John (1987) "Optimal Replacement of GMC Bus Engines: An Empirical Model of Harold Zurcher," technical report.

Schmutz, Benoît and Modibo Sidibé (2019) "Frictional Labour Mobility," *The Review of Economic Studies*, 86, 1779–1826, 10.1093/restud/rdy056.

Shimer, Robert (2012) "Reassessing the ins and outs of unemployment," *Review of Economic Dynamics*, 15, 127–148, 10.1016/j.red.2012.02.001.